

Challenges in KNN Classification

Shichao Zhang¹, Senior Member, IEEE

Abstract—The KNN algorithm is one of the most popular data mining algorithms. It has been widely and successfully applied to data analysis applications across a variety of research topics in computer science. This paper illustrates that, despite its success, there remain many challenges in KNN classification, including K computation, nearest neighbor selection, nearest neighbor search and classification rules. Having established these issues, recent approaches to their resolution are examined in more detail, thereby providing a potential roadmap for ongoing KNN-related research, as well as some new classification rules regarding how to tackle the issue of training sample imbalance. To evaluate the proposed approaches, some experiments were conducted with 15 UCI benchmark datasets.

Index Terms—Data mining, lazy learning, KNN classification, classification rule

1 INTRODUCTION

N (nearest neighbor) classification is an efficient solution to approximation, which was first proposed as a nonparametric discrimination in statistics [1]. However, it has long suffered from the key issue of overfitting [2], [3]. k nearest neighbor (KNN) classification was also advocated by Fix and Hodges [1] as a possible solution to this issue. Here, K objects are found in a training dataset that are closest to a test object/data. A label is then assigned according to the predominance of the majority class in this neighborhood. This is the standard prediction approach to KNN classification, known as the majority rule.

KNN classification has the remarkable property that, under very mild conditions, the error rate of a KNN algorithm tends towards being Bayes optimal as the sample size tends towards infinity [4]. For any data analysis application, if establishing a model with some training dataset is proving troublesome, it is likely that a KNN algorithm will provide the best solution [5]. As a result, KNN algorithms have been widely used in research and are considered to being one of the top-10 data mining algorithms [6]. In this era of big data, KNN approaches provide a particularly efficient way of identifying useful patterns and developing case-based reasoning algorithms for artificial intelligence (AI) [7], [8].

As with other classification intelligences, KNN classification is a two-phase procedure: model training and test data prediction. In the training phase, KNN only involves finding a suitable K for a given training dataset [9]. The most common method for this is the cross-validation. In the prediction phase, the first step is a search for K data points in the training dataset that are most relevant to a query (test data/sample). Without other information, the K most relevant data points are taken to be the K nearest neighbors of

the test data within the training dataset. After this, a prediction is made on the basis of the class of test data that most frequently occurs amongst the K neighbors. This is referred to as the majority rule (which is similar to the Bayesian rule). From the above procedure of KNN classification, it indicates that there are mainly four challenging issues, K computation, nearest neighbour selection, nearest neighbour search, and classification rule.

It can be very difficult to set a suitable K for a given training dataset. Training samples often have different distributions in the sample space, which can lead to there being no obviously suitable K for the whole training sample space. This has resulted in two research directions as follows. One is to set different K values to different sample subspaces [10], [11]. Another is to set different K values to different test samples [12]. From Zhang *et al.*, the efficiency is pretty good when clustering the sample space to 3-5 subspaces [11]. It is time-consuming to set different K values to different test samples.

Nearest neighbor selection has been studied extensively. It is really a procedure of determining a proximity measure. Much of the related research has focused on constructing distance functions for measuring the proximity. Song, *et al.* proposed two KNN methods for measures the informativeness of test data, Locally Informative-KNN (LI-KNN) and Globally Informative-KNN (GI-KNN) [13]. Each of them is as a query-based distance metric to measure the closeness between objects. However, no distance function has yet been identified that is suitable for all training samples regardless of their distribution. In other words, there remains a need for distance function for the selection of the K nearest neighbor points that can work effectively across most training samples. On the other hand, feature selection is useful to choosing the K nearest neighbors [14]. This can be a lazy procedure which depends on data mining tasks.

Nearest neighbor search is particularly challenging and remains unsolved because it is a complete sample space search when looking for all the K nearest neighbors for each test data. As a result, KNN classification is often referred to as a lazy data mining method. There are some efforts devoted to resolving the problem of how to undertake a truly effective nearest neighbor search. From the lately

• The author is with the College of Computer Science and Technology, Central South University, Changsha 410083, PR China.
E-mail: zhangsc@csu.edu.cn.

Manuscript received 21 Aug. 2019; revised 21 Dec. 2020; accepted 31 Dec. 2020.
Date of publication 5 Jan. 2021; date of current version 12 Sept. 2022.

(Corresponding author: Shichao Zhang.)

Recommended for acceptance by Y. Zhang.

Digital Object Identifier no. 10.1109/TKDE.2021.3049250